

Multivariate bias correction improves inter-station dependence in multi-site daily precipitation without altering single-site performance: an evaluation of MBCn against quantile delta mapping over a coastal gauge network in Thailand

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Abstract

Bias correction of climate-model precipitation is usually evaluated station by station, although distributed hydrological applications also require realistic dependence among gauges. We compared N-dimensional multivariate bias correction (MBCn), quantile delta mapping (QDM), and uncorrected output for 12 daily rain gauges on the western Gulf of Thailand coast and seven CMIP6 models. Methods were calibrated on 1981-2000 and evaluated independently on 2001-2014. Because the final MBCn step reinstates QDM values at each gauge, both methods produced identical results for six marginal diagnostics across all 84 gauge-model combinations. MBCn improved all six dependence diagnostics. Relative to QDM, it reduced energy distance from 0.852 to 0.571 and Spearman correlation-matrix RMSE from 0.286 to 0.187; all seven models improved on every diagnostic (one-sided exact Wilcoxon $p = 0.0078$; rank-biserial effect size = 1.00). Most dependence gains occurred within 10-15 iterations, results were insensitive to random seed, and projected wet-day quantile changes remained comparable to QDM. QDM is therefore sufficient when only station-level distributions are required, whereas MBCn is better supported when spatial dependence matters.

Keywords: multivariate bias correction; MBCn; inter-station dependence; multi-site precipitation; CMIP6

1. Introduction

Global climate models remain too coarse for direct use at catchment and station scales, and their daily precipitation contains systematic errors in both occurrence and intensity [1]. Distribution-based bias correction is therefore standard in hydrological climate-impact studies [2], [3]. Quantile delta mapping (QDM) is widely used because it adjusts the full distribution at a site while preserving the model-projected relative change at each quantile [4]. This is usually sufficient when the analysis treats each gauge independently.

QDM does not directly control relationships among gauges because each site is corrected independently. This limitation matters when corrected fields drive distributed hydrological models, joint flood analyses, or assessments of spatially extensive droughts and storms. Accurate marginal distributions can coexist with an unrealistic joint structure, and such errors propagate into analyses of compound and spatially concurrent extremes [5].

Multivariate methods address the joint distribution explicitly. Examples include rank-based resampling [6], optimal transport [7], and N-dimensional multivariate bias correction (MBCn) [8]. MBCn repeatedly rotates the multistation field, quantile-maps each rotated component to the corresponding observations, and reverses the rotation; a final step reinstates the QDM marginal values at each gauge. Comparative studies show that multivariate adjustment can improve dependence while introducing method-specific trade-offs [9], and regional applications confirm the relevance of evaluating spatial as well as marginal behaviour [10].

The final rank-substitution step determines how QDM and MBCn should be compared. Because MBCn restores the QDM values at every gauge, the two methods must be identical for distribution-based single-site diagnostics. Their added value can therefore be judged only with diagnostics that measure dependence among gauges.

The spatial organisation of rainfall extremes is both consequential and uncertain under climate change [11]. CMIP6 studies have evaluated precipitation change across mainland Southeast Asia [12]–[14] and Thailand [15]–[18], while recent work has compared correction methods in multistation settings [19]. However, few studies combine an independent validation period with several complementary dependence diagnostics in one gauge-network experiment. We therefore ask: once single-site performance is fixed by construction, how much does MBCn improve the joint behaviour of a coastal gauge network, and does it compromise numerical stability or the projected quantile-change signal?

2. Materials and Methods

2.1 Study area and rain-gauge network

Twelve daily rain gauges in Phetchaburi and Prachuap Khiri Khan provinces, on the western shore of the Gulf of Thailand, provide the observations (Table 1 and Figure 1). Each record is complete from 1 January 1981 to 31 December 2014. Mean annual precipitation ranges from approximately 944 to 1,376 mm, and wet-day frequency varies markedly across the network. This spatial contrast provides a demanding test of whether correction preserves local distributions while improving network dependence.

Coordinates and elevations were obtained from the project metadata archive. Figure 1 locates the gauges within the provincial boundary. The background shading is an inverse-distance interpolation of mean annual precipitation for orientation only; it is not used in any analysis.

Station	Latitude (°N)	Longitude (°E)	Elevation (m)	Mean annual rainfall (mm)	Wet-day frequency (%)
500001	12.370	99.950	8	983.7	22.23
500002	11.180	99.550	0	1,376.2	33.73
500003	12.020	99.900	0	1,100.0	22.93
500004	11.470	99.650	0	1,106.0	26.34
500005	11.600	99.680	6	1,147.2	47.96
500006	11.620	99.670	24	1,203.1	49.77
500007	11.900	99.780	24	1,133.6	48.66
500008	12.530	99.550	128	1,206.3	62.63
500009	12.220	99.870	13	1,176.5	60.15
500201	11.830	99.830	64	1,108.2	24.70
500202	12.590	99.960	0	944.1	21.53
500301	12.580	99.730	106	1,064.8	25.33

Table 1. Characteristics of the twelve rain gauges, 1981–2014.

Coordinates (WGS84) and elevations are from the project metadata archive; rainfall statistics are computed from the gap-free observed daily record. A wet day is defined throughout as at least 1 mm of rainfall.

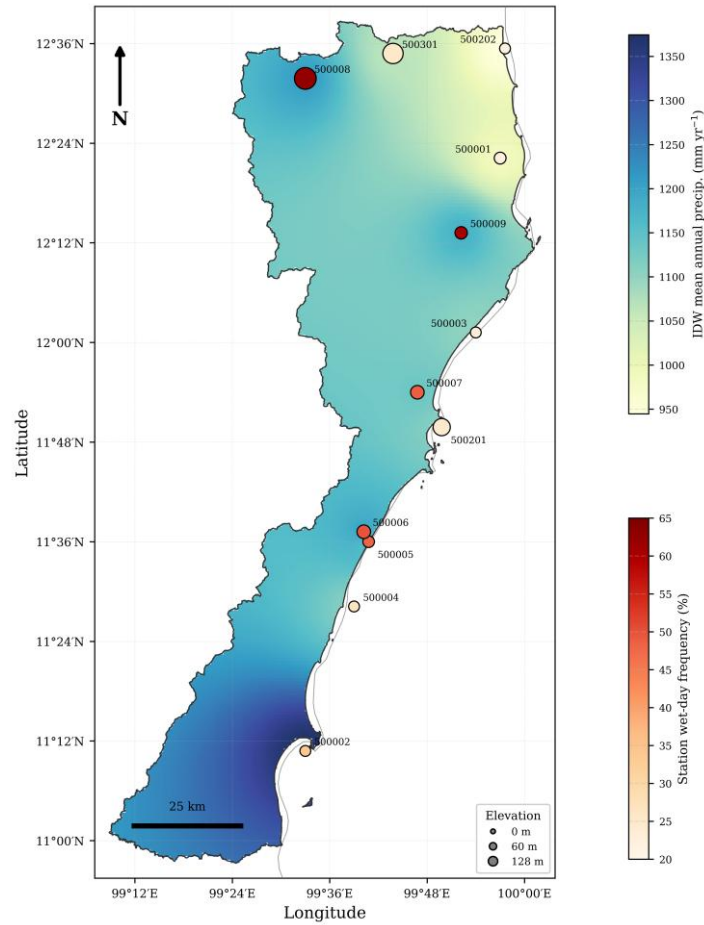


Figure 1. The twelve-gauge network on the Phetchaburi–Prachuap Khiri Khan coast. Marker size scales with elevation and fill with wet-day frequency. The shaded background is an inverse-distance surface of mean annual rainfall, shown for orientation only and clipped to the provincial boundary (reprojected from UTM Zone 47N).

2.2 Climate models and experimental design

Seven CMIP6 models supplied daily precipitation for the gauge locations (Table 2). Ensemble-member and grid labels were verified against the archive filenames. CESM2 used r1i1p1f1; the other six models used r1i1p1f1.

A single split was used for every method: 1981–2000 for calibration and 2001–2014 for independent validation. Validation observations were not used during fitting. All comparisons were restricted to the same 12 gauges.

Model	Institution	Ensemble member	Grid label
ACCESS-ESM1-5	CSIRO, Australia	r1i1p1f1	gn
CanESM5	CCCma, Canada	r1i1p1f1	gn
CESM2	NCAR, United States	r1i1p1f1	gn
EC-Earth3	EC-Earth Consortium, Europe	r1i1p1f1	gr
FGOALS-g3	CAS, China	r1i1p1f1	gn
MIROC6	MIROC, Japan	r1i1p1f1	gn
MRI-ESM2-0	MRI, Japan	r1i1p1f1	gn

Table 2. The seven CMIP6 models and their configuration.

All seven models provide daily precipitation for the historical experiment (1981–2014) and for both SSP2–4.5 and SSP5–8.5. Labels were verified against the archive filenames. gn = native model grid; gr = regrided.

2.3 Bias-correction methods

QDM was applied separately to each gauge and calendar month. Observed and simulated quantile functions were estimated in the calibration period, and relative model-projected changes were preserved at each quantile [4]. MBCn began from the QDM-corrected field and applied iterative multivariate adjustment [8]. Each of 25 passes used a seeded random orthogonal rotation across the 12 station dimensions, mapped each rotated component to the corresponding rotated observations, and reversed the rotation. Final rank substitution restored the QDM values at every station, retaining QDM marginal distributions while adjusting inter-station ranks (Figure 2). Related multistation comparisons also highlight the need to evaluate correction methods beyond marginal performance [19].

All QDM mappings, rotations, and rotated-space mappings were fitted using 1981-2000 data only. The 2001-2014 observations were reserved for evaluation. A wet day was defined as precipitation of at least 1 mm. Thresholds for extreme-event diagnostics were estimated separately from each series' wet days so that these diagnostics primarily reflected dependence rather than residual intensity differences.

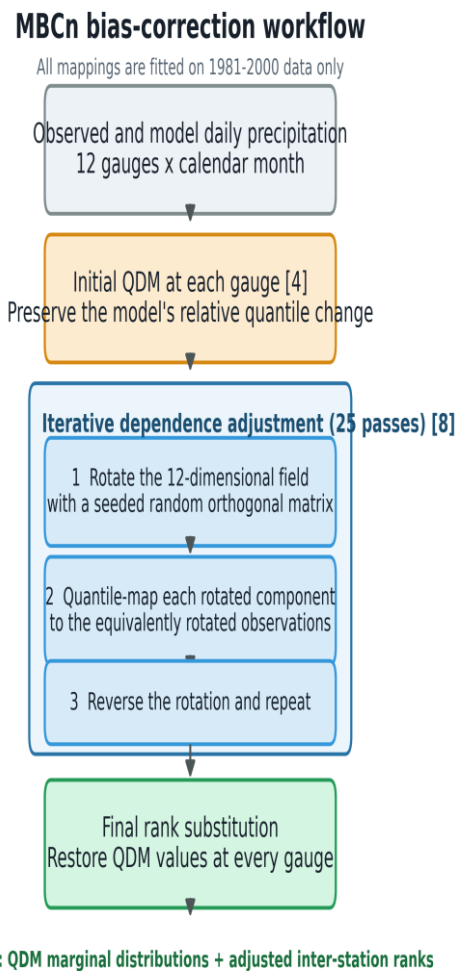


Figure 2. MBCn workflow. QDM [4] first establishes each gauge's marginal distribution; the iterative rotation, mapping, and inverse-rotation stage [8] adjusts inter-station dependence; final rank substitution restores the QDM values. All transformations are fitted using calibration data only.

2.4 Single-site diagnostics

Because free-running climate simulations are not synchronized with observed daily weather, single-site performance was assessed distributionally. Six diagnostics were computed for each of 84 gauge-model combinations: percentage bias in total precipitation, wet-day-frequency error, the ratio of

simulated to observed wet-day standard deviation, Perkins skill score [20], Kolmogorov-Smirnov distance, and RMSE between matched wet-day quantiles. Day-matched criteria were not used because they would penalize timing that free-running simulations are not designed to reproduce.

2.5 Dependence diagnostics and statistical testing

Six diagnostics evaluated joint behaviour. Energy distance compared the 12-dimensional daily precipitation vectors with observations [21] using a fixed-seed subsample of at most 1,500 validation days. Three metrics measured RMSE across the 66 off-diagonal station pairs for Spearman correlation, Pearson correlation, and binary wet-day correlation. Two metrics targeted concurrent extremes: RMSE of pairwise joint exceedance above each series' 95th wet-day percentile and the Wasserstein-1 distance between distributions of daily network wet extent. Each diagnostic was computed separately for each climate model.

Seven paired model-level differences between MBCn and QDM were summarized with a one-sided exact Wilcoxon signed-rank test, the matched-pairs rank-biserial effect size, and the number of models favouring MBCn. The test describes consistency within this ensemble rather than inference to a population of climate models. Holm-Bonferroni adjustment was applied to the prespecified family of Spearman, Pearson, and wet-day-correlation endpoints; the other diagnostics were treated as exploratory.

2.6 Robustness checks

Robustness was evaluated by varying the number of MBCn passes from 0 (QDM) to 100 and repeating the default 25-pass correction with 20 random seeds. Fidelity to the model-projected signal was evaluated by comparing raw and corrected percentage changes in five wet-day quantiles for every gauge, model, pathway, and future period. This check addresses quantile changes only and does not validate the projections themselves.

3. Results and Discussion

3.1 Single-site performance is identical between the two corrections

The raw output produced rain too frequently and underestimated wet-day variability (Table 3 and Figure 3). After correction, Kolmogorov-Smirnov distance decreased from 0.221 to 0.101 and Perkins skill score increased from 0.82 to 0.91. Both corrections retained an approximately 17% dry bias in total precipitation and a wet-day-frequency bias of about -6 percentage points; this residual bias should be considered in water-balance applications.

QDM and MBCn were identical to three decimal places for every marginal score across all 84 gauge-model combinations. This confirms the expected algorithmic behaviour: final rank substitution returns the same set of daily QDM values at each gauge. The multivariate stage can therefore affect only how values are arranged jointly across the network, not their station-level distributions.

Method	PBIAS (%)	Wet-day frequency error (pp)	SD ratio (-)	PSS (-)	KS distance (-)	Quantile RMSE (mm d ⁻¹)
RAW	1.4 ± 34.7	10.6 ± 17.2	0.69 ± 0.49	0.82 ± 0.10	0.221 ± 0.104	7.83 ± 4.16
QDM	-17.2 ± 20.8	-6.4 ± 12.3	0.98 ± 0.36	0.91 ± 0.05	0.101 ± 0.060	4.08 ± 3.05
MBCn	-17.2 ± 20.8	-6.4 ± 12.3	0.98 ± 0.36	0.91 ± 0.05	0.101 ± 0.060	4.08 ± 3.05

Table 3. Single-site skill over the withheld validation period (mean ± SD, n = 84 gauge-model pairs).

pp = percentage points; *PSS* = Perkins skill score; *KS* = Kolmogorov-Smirnov; *SD ratio* is the corrected-to-observed ratio of the wet-day standard deviation. *QDM* and *MBCn* coincide on every score because *MBCn* reinstates the *QDM* value at each gauge.

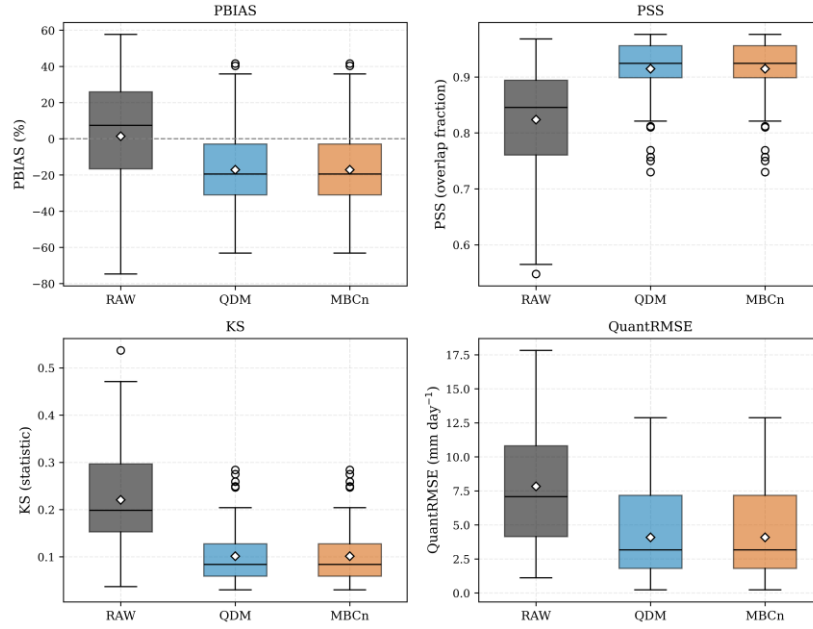


Figure 3. Single-site skill for RAW, QDM, and MBCn over the withheld validation period, one point per gauge–model pair ($n = 84$). Diamonds mark the mean. The QDM and MBCn distributions coincide exactly, by construction.

3.2 MBCn improves inter-station dependence consistently across models

MBCn outperformed QDM for every dependence diagnostic (Table 4). Energy distance decreased from 0.852 for QDM to 0.571 for MBCn. Spearman correlation-matrix RMSE decreased from 0.484 in raw output to 0.286 with QDM and to 0.187 with MBCn. Pearson correlation, wet-day correlation, joint 95th-percentile exceedance, and wet-extent distribution showed the same ordering.

Every model improved under MBCn for all six diagnostics (Table 5). With seven concordant non-zero paired differences, the one-sided exact Wilcoxon p -value was 0.0078, the smallest attainable value at $n = 7$, and the matched-pairs rank-biserial effect size was 1.00. Holm-Bonferroni-adjusted p -values for the three prespecified correlation endpoints were 0.0234. These values summarize a unanimous within-ensemble pattern; the models are not a random sample, and the six diagnostics are related views of the same dependence improvement.

The consistency across models is empirical rather than guaranteed. Figure 4 shows the Spearman matrices for MIROC6, selected because its MBCn error was closest to the ensemble median. Raw output imposed uniformly high correlation; QDM reduced it only partly; and MBCn recovered more of the observed spatial contrast. Figure 5 shows all seven paired model results, each of which favours MBCn.

Figure 6 relates pairwise Spearman correlation to gauge separation. Observed correlation declines modestly with distance, whereas raw output remains near 0.96 across most pairs. QDM reduces this over-coherence, and MBCn most closely follows the observed distance relationship. Coarse-grid extraction contributes to the raw over-coherence when gauges share or occupy adjacent grid cells. The binned relationship and bootstrap bands are descriptive because the 66 gauge pairs are not independent.

Method	Energy distance	Spearman RMSE	Pearson RMSE	Wet-occurrence RMSE	Joint q95 RMSE	Wet-extent W_1
RAW	1.283	0.484	0.488	0.476	0.0150	0.174
QDM	0.852	0.286	0.374	0.273	0.0054	0.107
MBCn	0.571	0.187	0.134	0.139	0.0029	0.079

Table 4. Dependence diagnostics against observations (seven-model mean, withheld validation period). All scores are negatively oriented, so lower is better.

Values are means of per-model scores. The energy distance is the broadest whole-distribution measure, computed on a fixed-seed random subsample of at most 1,500 days; the correlation-matrix RMSEs, the joint-exceedance RMSE and the wet-extent Wasserstein-1 distance (W_1) are structure diagnostics. Joint exceedance uses each series' own wet-day ninety-fifth percentile and wet extent the 1 mm threshold. MBCn is closest to observations on every diagnostic. All quantities are dimensionless.

Diagnostic	Models favouring MBCn	W	p	Holm-adjusted p	Effect size
Energy distance (secondary)	7 / 7	28.0	0.0078	—	1.00
Spearman RMSE (primary)	7 / 7	28.0	0.0078	0.0234	1.00
Pearson RMSE	7 / 7	28.0	0.0078	0.0234	1.00
Wet-day co-occurrence RMSE	7 / 7	28.0	0.0078	0.0234	1.00
Joint q95 exceedance RMSE (secondary)	7 / 7	28.0	0.0078	—	1.00
Wet-extent W_1 (secondary)	7 / 7	28.0	0.0078	—	1.00

Table 5. Paired comparison of MBCn against QDM across the seven models (one-sided Wilcoxon signed-rank test; Holm–Bonferroni adjustment within the correlation-endpoint family).

Every model improves under MBCn on every diagnostic, yielding the maximum matched-pairs rank-biserial effect size and the smallest p attainable at $n = 7$. The Holm adjustment covers the three pre-declared correlation endpoints; the remaining diagnostics are exploratory and reported unadjusted.

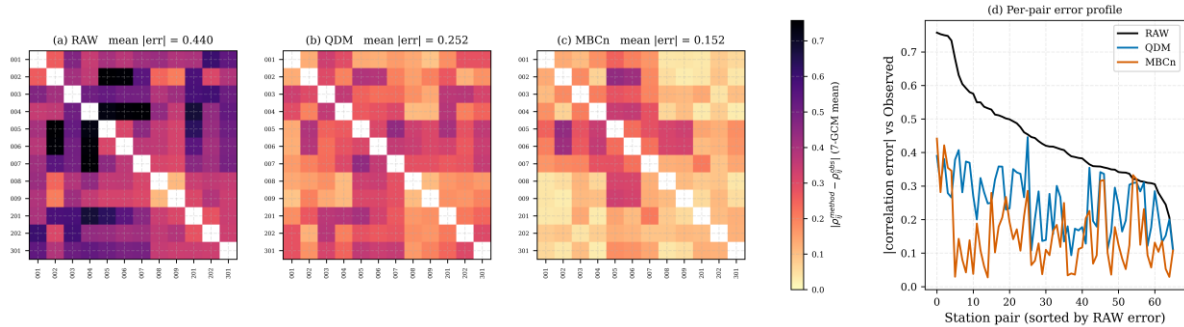


Figure 4. Inter-station Spearman correlation for the median-performing model (MIROC6, selected before inspection). RAW imposes a near-uniform high correlation, QDM corrects it only partially, and MBCn recovers a texture close to the observed. Panel (d) shows the sorted per-pair error.

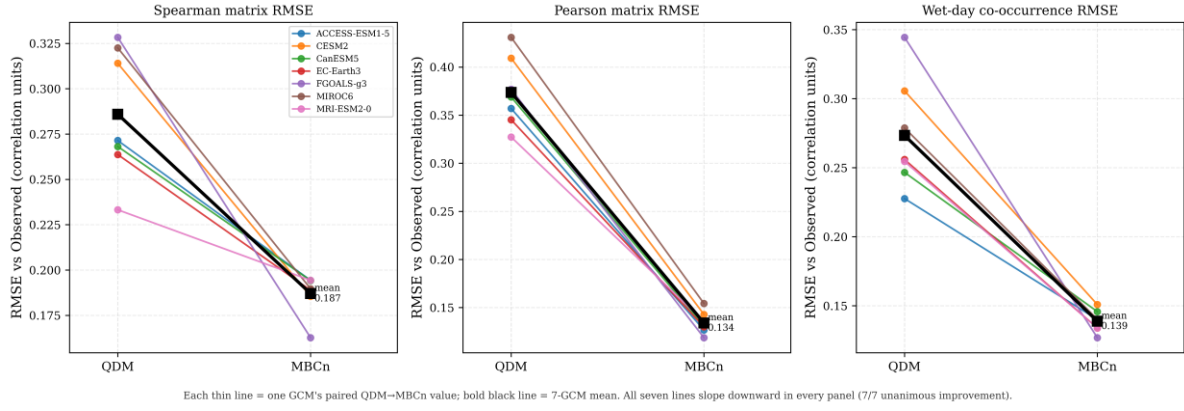


Figure 5. Paired QDM→MBCn dependence error for each of the seven models (thin lines) and the ensemble mean (bold), for three diagnostics. Every line slopes downward, the graphical form of the 7/7 result in Table 5.

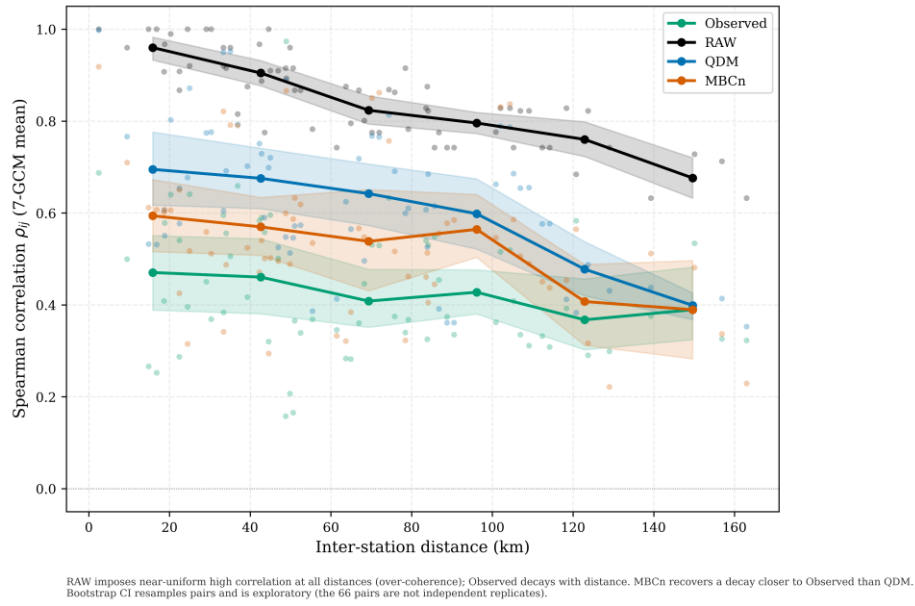


Figure 6. Inter-station Spearman correlation versus gauge separation. MBCn follows the observed distance relationship more closely than QDM, whereas raw output remains uniformly high. Bootstrap bands are exploratory because gauge pairs are not independent.

3.3 Spatial organisation of concurrent extremes

Two event-oriented diagnostics complement the correlation analysis. Figure 7 compares observed and simulated probabilities that both gauges in a pair exceed their own 95th wet-day percentiles on the same date. Raw output is widely dispersed, QDM retains a positive bias, and MBCn lies closest to the 1:1 line without a systematic tilt.

Figure 8 shows the pooled distribution of the number of gauges wet on each day. MBCn is closest to the observed distribution, consistent with the per-model wet-extent scores in Tables 4 and 5. The inset values in Figure 8 are calculated from the pooled distributions, whereas Table 4 reports the mean of seven separately calculated model-level Wasserstein-1 distances; the two summaries therefore need not be numerically identical.

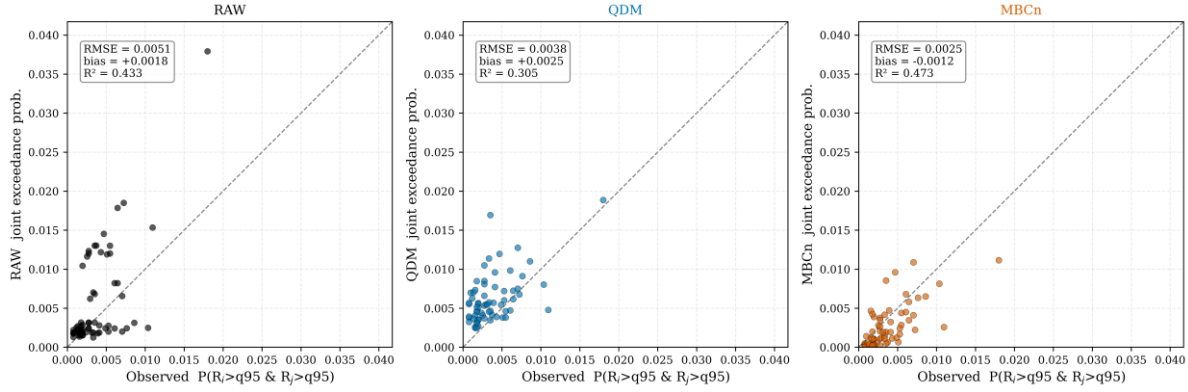


Figure 7. Pairwise probability that both gauges exceed their ninety-fifth wet-day percentile on the same day, observed against corrected (seven-model mean). MBCn lies closest to the 1:1 line and is the only method without a systematic tilt.

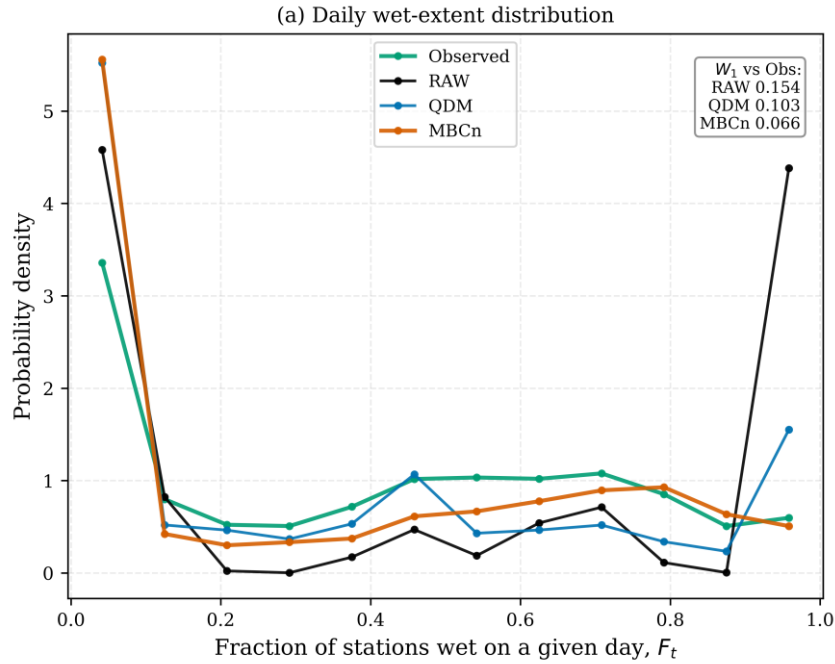


Figure 8. Pooled distribution of the daily number of simultaneously wet gauges for observations and each method. Inset Wasserstein-1 values are computed from the pooled distributions; Tables 4 and 5 use separately computed per-model scores.

3.4 Numerical behaviour and fidelity to the projected change signal

Figure 9 shows the convergence analysis for three representative climate models. Most improvement occurred within 10-15 passes; diagnostics then oscillated within a narrow range, placing the 25-pass default on the plateau. Across 20 random seeds, correlation-based scores changed little (coefficient of variation approximately 4%). The larger spread in energy distance reflects its fixed-size random subsampling and does not appear in the full-sample correlation diagnostics.

Projected wet-day quantile changes were similarly preserved by both methods (Figure 10). Across all gauges, models, pathways, and future periods, mean absolute deviation from the raw change signal was 9.4 percentage points for QDM and 9.2 for MBCn; the corresponding R-squared values were 0.485 and 0.465. Thus, MBCn achieved its dependence gains without a material additional loss of the QDM change-preservation property [4]. This check does not cover sequence-dependent statistics, such as wet- and dry-spell lengths, which MBCn may alter.

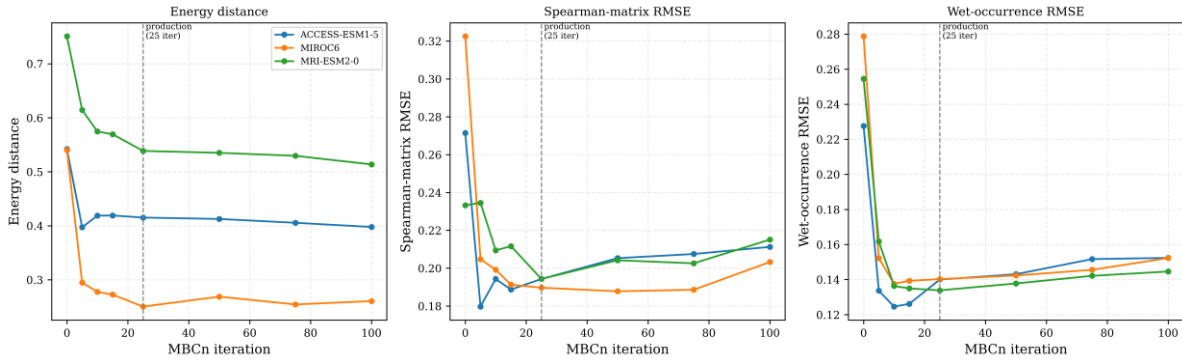


Figure 9. Dependence diagnostics versus MBCn iteration count for three representative climate models; iteration 0 is QDM. Most improvement occurs within approximately 10-15 passes, placing the default of 25 on the plateau.

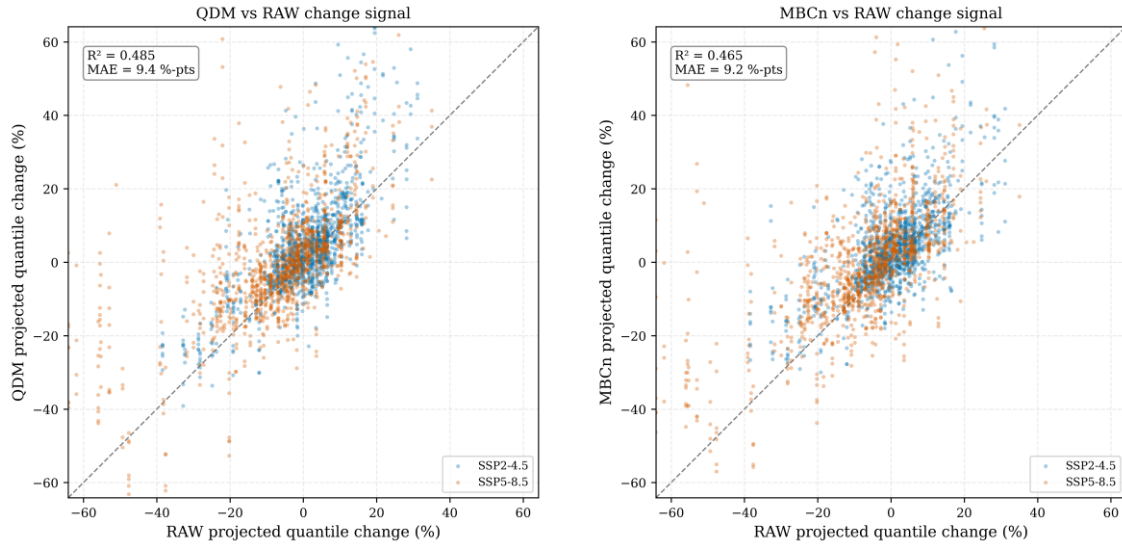


Figure 10. Raw versus corrected projected change in wet-day quantiles for every gauge, model, pathway, and future period. QDM and MBCn preserve the model-projected quantile-change signal to a similar degree.

3.5 Implications and limitations

The results support a context-specific choice rather than a universal ranking. Recent comparisons likewise show that no bias-correction method dominates every temporal and dependence property [22]; method selection should follow the structure required by the downstream application. QDM is sufficient and less computationally demanding when only single-site distributions are required, including many intensity-duration-frequency applications in Thailand [23]. MBCn is better supported when gauges are used jointly in distributed hydrological modelling, spatial hazard mapping, or multivariate risk analysis.

More broadly, station-wise evaluation cannot distinguish methods that share marginal distributions but differ in joint behaviour. At least one dependence diagnostic should accompany conventional marginal scores whenever corrected variables or locations will be analysed together.

Four limitations bound the conclusions. One realization per model does not represent internal variability. The single coastal network may not represent inland or more strongly monsoonal settings. The stationary calibration assumes that historical dependence remains an appropriate target under future climate. Finally, the residual dry bias in both corrections limits water-balance applications. These limitations constrain generalization but do not affect the paired comparison, for which both methods used identical inputs and periods.

4. Conclusion

Across 12 coastal gauges and seven CMIP6 models, QDM and MBCn were identical for all six station-level diagnostics because MBCn restored the QDM marginal values. In contrast, MBCn improved all six dependence diagnostics for every model, reducing energy distance and correlation-structure errors while converging rapidly, remaining insensitive to random seed, and preserving projected wet-day quantile changes comparably to QDM. QDM therefore remains appropriate when only marginal fidelity is needed; MBCn is the better-supported choice when inter-station dependence is part of the scientific or hydrological question. Dependence diagnostics should become routine whenever corrected locations are used jointly. Evaluation across larger, topographically diverse networks and multi-realization ensembles is the next step for testing the generality of these findings.

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Author Contributions

Surasit Punyawansiri: conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing - original draft, writing - review and editing, visualization, supervision, and project administration. The author has read and approved the submitted manuscript.

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Conflicts of Interest

The author declares no conflict of interest.

Data and Code Availability

The daily rain-gauge records are available from the Royal Irrigation Department subject to its data-sharing conditions. CMIP6 output is publicly available through the Earth System Grid Federation. Analysis code and underlying numerical data for the figures are available from the corresponding author on reasonable request.

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